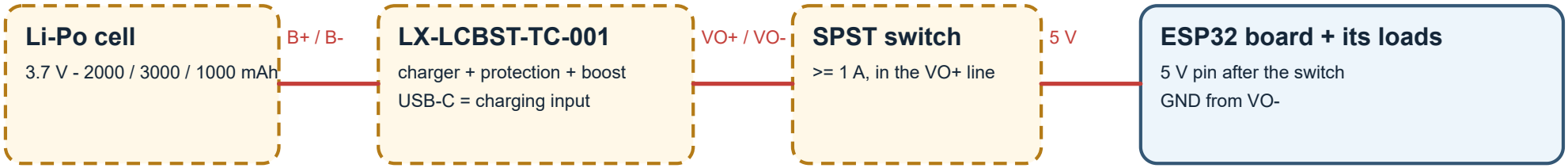
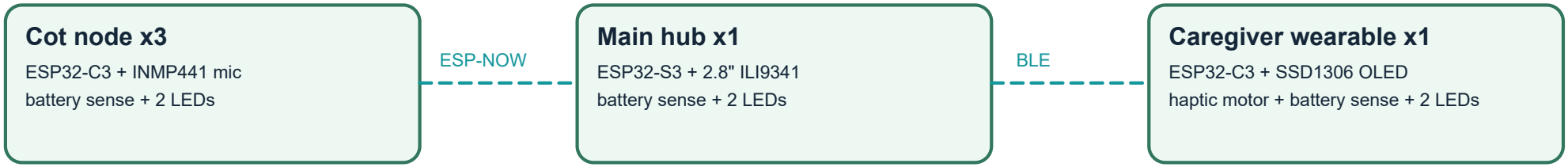
1 / CryNet system - power chain and links

Every unit has its own battery and its own ground. No wire ever joins two units.



One chain in all four units - only the loads differ:



Rules every unit follows:

1 - Switch OFF before charging. The cell still charges on B+/B-.

2 - Set the boost trimpot to 5.1 V with a meter before any board.

3 - Never power a board from USB and the boost at the same time.

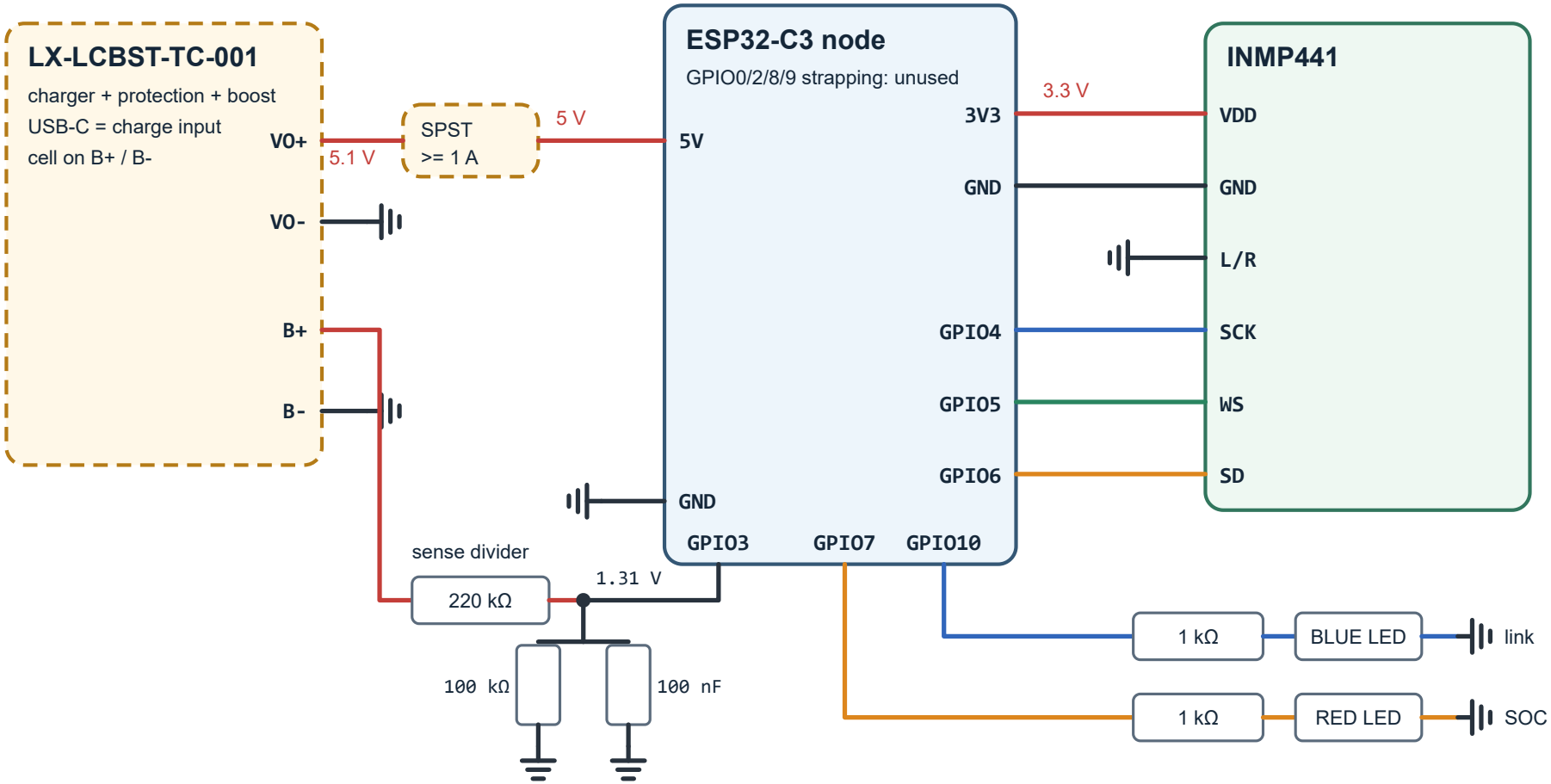
Dashed teal = wireless link, no wire. Solid = a real conductor.

> Proposed, bench-validation pending: the whole battery chain, the wearables and the LEDs. The node and hub signal wiring is the verified part.

No signal wire exists between the nodes and the hub - the link is wireless. Each unit is electrically independent; never join their batteries or grounds. Read with SYSTEM_WIRING.md sections 5 and 6.

2 / Cot node - complete wiring (one per node, x3)

ESP32-C3 SuperMini + INMP441 + LX-LCBST + switch + battery-sense divider + 2 LEDs



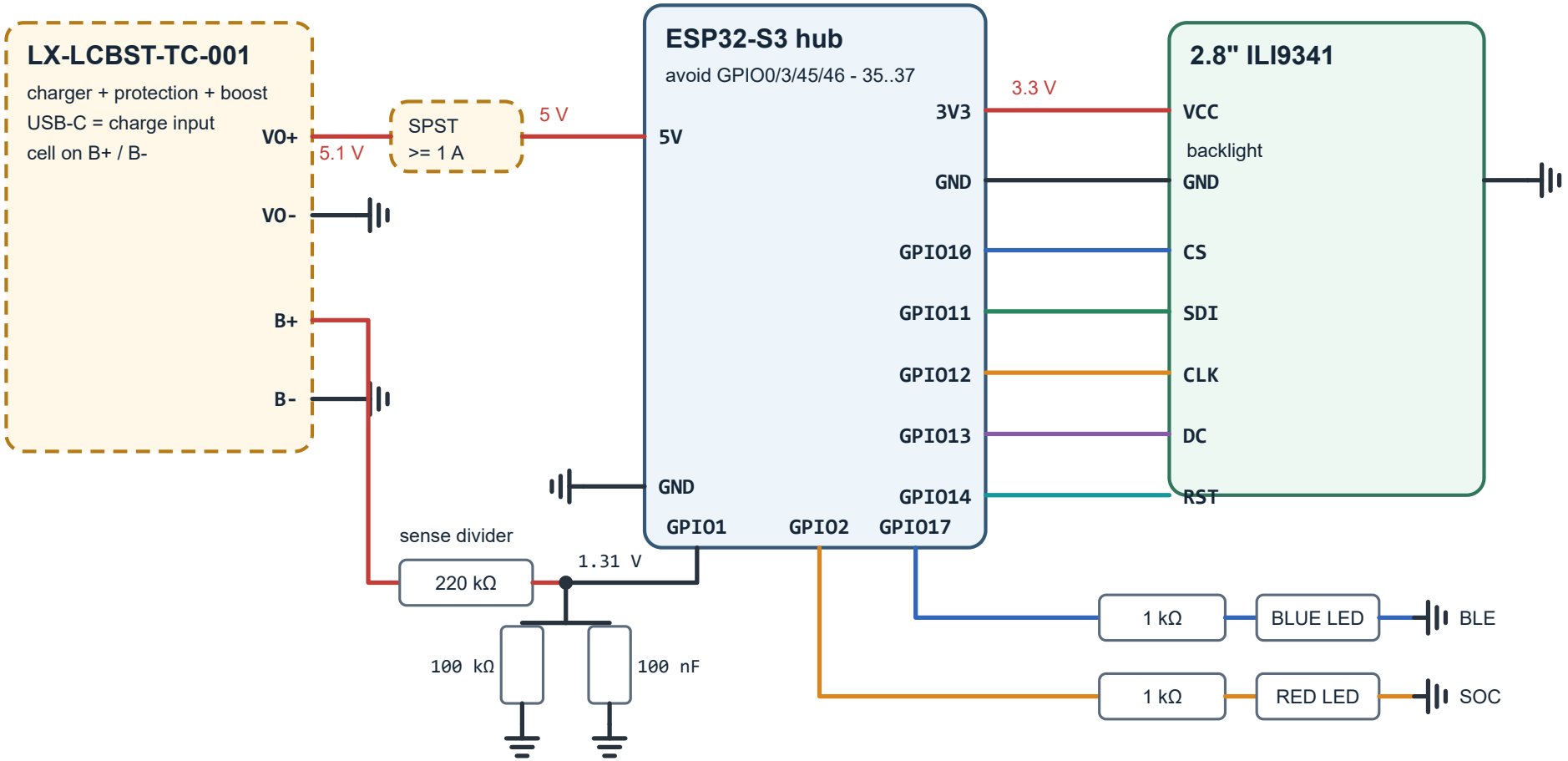
Verified on the live system: mic wiring (3V3 / GND / L/R→GND, GPIO4 SCK, GPIO5 WS, GPIO6 SD) and the node firmware. GPIO1 stays spare.

Proposed, needs bench validation: the whole battery chain, the divider and the LEDs. The divider reads about 13 uA. RED LED = SOC (off / slow / fast), BLUE LED = link (ACK-driven). Firmware for the divider and the LEDs does not exist yet - branch work.

Every ⏚ symbol is the same net: the module's VO- (battery negative through the module's protection FET). Wires join only where a dot is drawn. Pin names, not header positions. Repeat identically for nodes 2 and 3.

3 / Main hub - complete wiring (x1)

ESP32-S3-DevKitC-1 N16R8 + 2.8" ILI9341 + LX-LCBST + switch + battery-sense divider + 2 LEDs



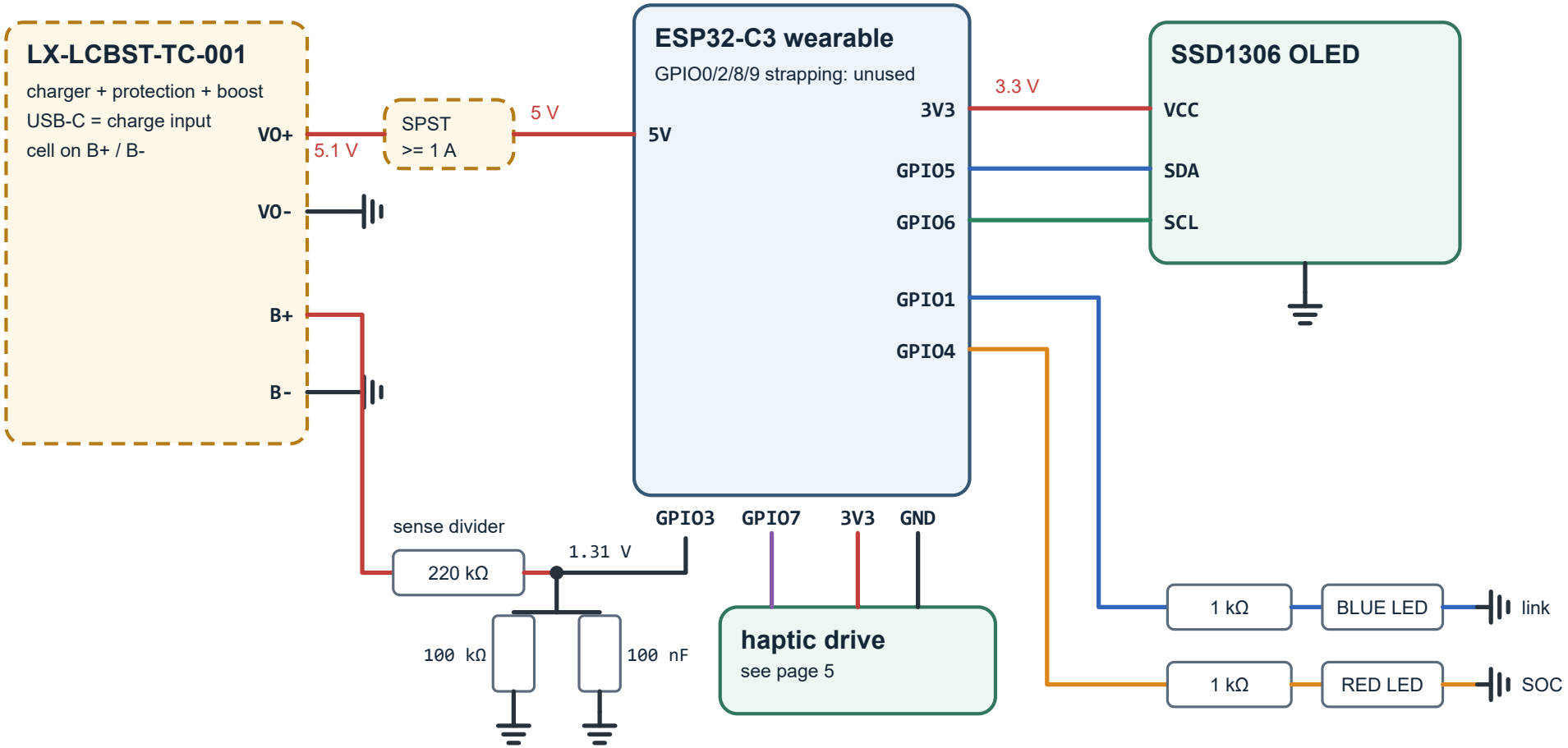
Verified on the live system: TFT signals (CS 10, MOSI 11, SCK 12, DC 13, RST 14, BL 4, VCC 3V3, GND) and the hub firmware.

Proposed, needs bench validation: the battery chain, the divider on GPIO1 (ADC1_CH0), BLUE LED on GPIO2 (BLE client connected) and RED LED on GPIO17 (SOC). The hub's peak current with TFT + radios is unmeasured - confirm the LX-LCBST holds 5.1 V during bursts before trusting it here.

GPIO35-37 are the Octal PSRAM bus - never wire them. The TFT wiring is the recorded working one for the existing module. Every ⏚ symbol is the same net: the module's VO-.

4 / Caregiver wearable - complete wiring (x1)

ESP32-C3 SuperMini + SSD1306 OLED + LX-LCBST + switch + battery-sense divider + 2 LEDs

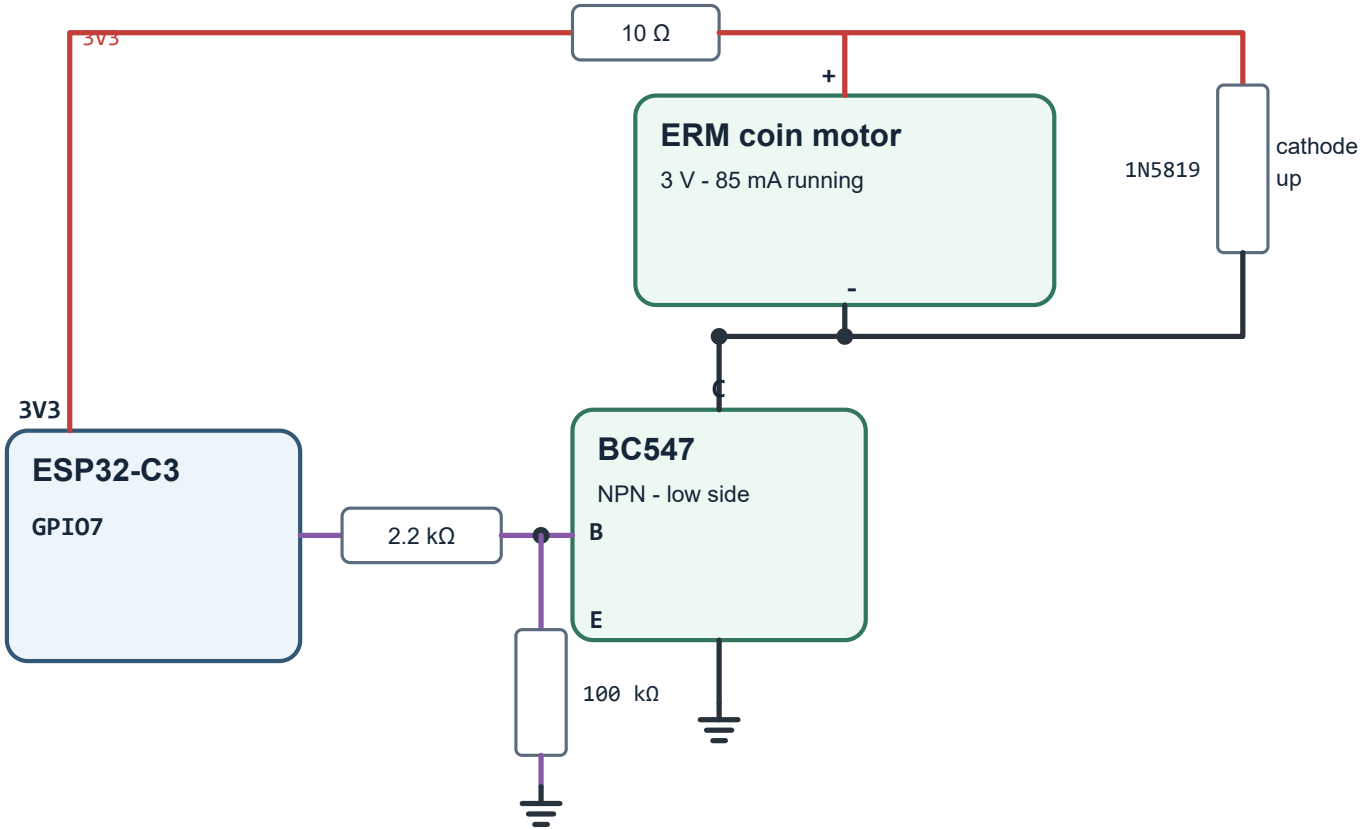


Proposed, nothing built: the OLED + haptic path. The OLED runs from the SuperMini 3V3 pin, never from 5 V - I2C pull-ups ride on VCC and 5 V there would push 5 V levels into the C3 GPIOs. The motor is on its own switch, drawn in full on page 5 - it is not powered from the 3V3 pin here. Divider on GPIO3; BLUE LED GPIO1 (BLE link), RED LED GPIO4 (SOC).

Every ⏚ symbol is the same net: the module's VO-. Wires join only where a dot is drawn. Pin names, not header positions. The haptic drive continues on page 5.

5 / Wearable haptic drive - BC547 low-side switch

One transistor, one coin motor, one flyback diode - no I2C driver module



Why these values

3 V coin ERM, typical: running 80-90 mA, start-up and stall 120-200 mA

BC547 is rated 100 mA, so the 10 Ω in the motor's + lead caps the stall current near 75 mA and keeps the transistor inside its rating.

Base: 2.2 kΩ gives 1.2 mA of base current. A BC547A (hFE >= 110) then drives 130 mA - saturated, no base-emitter leak.

100 kΩ base-emitter holds the motor off while the C3 boots, when GPIO7 floats.

1N5819 catches the back-EMF when the transistor switches off.

Firmware drives GPIO7 high for about 250 ms per cry alert and low the rest of the time.

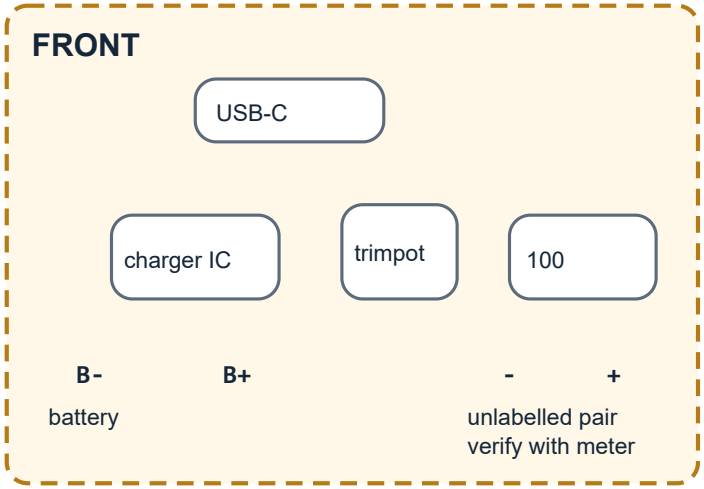
Check the motor before soldering: if its stall current exceeds 100 mA, the 10 Ω resistor must grow (22 Ω for a 200 mA motor) or the BC547 must be swapped for a 2N2222A. Measure the motor's stall current with a bench supply and a meter in series - do not guess.

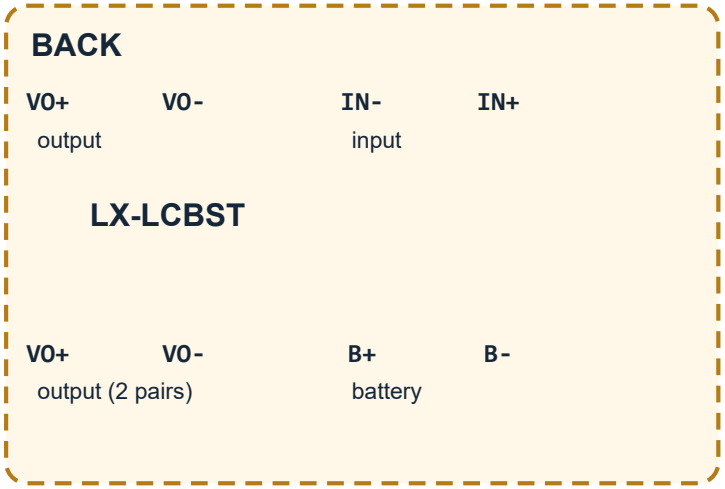
One buzz per cry: the alert notify characteristic carries the event id, and the sketch only fires haptic_trigger() when the alert type changes to a cry, so a louder cot joining a cry does not re-buzz.

The BC547 emitter is the motor's ground: the motor's negative terminal goes to the collector, its positive terminal to 3V3. Keep the flyback diode across the motor itself, cathode to the + terminal.

6 / LX-LCBST-TC-001 - pad map and bench procedure

From the user's front-and-back photograph - Chinese silk text translated





Bench procedure, before any ESP32 is connected:

1 - Cell on B+/B-: turn the trimpot until VO+/VO- reads 5.1 V on a meter, no load.

2 - USB-C in: measure the charge current with a series meter - the cell must allow it.

3 - Confirm the charge ends at 4.2 V.

4 - Switch off: confirm the load is dead and measure the battery drain (boost quiescent unknown).

5 - Run one node and compare it with the USB baseline.

No T pad on this board - the listing's T-shorting notes belong to other variants. Never short anything.

Unknown until measured: the actual charge current (the program resistor is unreadable in the photo), the boost quiescent drain when off, the protection thresholds, and what the front - / + pair connects to. Buy ONE module for validation; only then buy the remaining three (plus the wearable's, pending the same check).